Common setup (run once)

```
pip install nltk
```

```
import nltk

# Download the NLTK datasets/models needed by the lab programs.
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('brown')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('universal_tagset')
nltk.download('conll2000')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')
```

Run each program in Jupyter Notebook, Google Colab, or VS Code with Python installed. Some programs download corpora on first run and need an internet connection.

4. Demonstrate the morphology of a word using an Add-Delete table

Aim: To illustrate how suffixes and prefixes can be added to or deleted from a word to derive related forms.

```
# Define the word transformations in an Add-Delete table.
# Format: original word, addition, deletion, resulting word.
morphology_table = [
    ('play', 's', '-', 'plays'),
    ('play', 'ed', '-', 'played'),
    ('play', 'ing', '-', 'playing'),
    ('playing', '-', 'ing', 'play'),
    ('played', '-', 'ed', 'play'),
    ('happy', 'ness', 'y', 'happiness'),
    ('unhappy', '-', 'un', 'happy')
]

# Print the table header.
print(f"{'Original':<12}{'Add':<10}{'Delete':<10}{'Result':<15}")
print("-" * 47)

# Display each morphological operation.
for original, add, delete, result in morphology_table:
    print(f"{original:<12}{add:<10}{delete:<10}{result:<15}")
```

Sample output

```
Original    Add       Delete    Result
-----------------------------------------------
play        s         -         plays
play        ed        -         played
play        ing       -         playing
playing     -         ing       play
played      -         ed        play
happy       ness      y         happiness
unhappy     -         un        happy
```

Explanation:

• **Add:** Adds a suffix or prefix.

• **Delete:** Removes an affix or replaces part of a word.

• For example, `play + ing = playing`.

• For `happy → happiness`, the final `y` is replaced by `i` before adding `ness`.

The table represents illustrative rules, not a full automatic morphological engine.